Quantum Scientific Computing (QSC2026)

Homework 1

Assigned: Due: _____

Instructions. Eight problems drawn from Lectures 1, 3, and 4 (foundations; Grover search; the quantum Fourier transform and phase estimation). Each is worth 10 points. Show your work: a correct number with no derivation earns little credit, and a clear argument with a small arithmetic slip earns most of it. You may use any result stated in the lecture notes, citing it by name. Collaboration is allowed on ideas, but write up your solutions independently.

Part I — Foundations (Lecture 1)

Problem 1. Entanglement and no-cloning. *(10 pts)* Work entirely with pure states.

(a) Compute $\mathrm{CNOT}\,|+\rangle\,|+\rangle$ and decide whether the result is a product state or entangled, justifying your answer.

(b) Exhibit a unitary U and a fixed "blank" state $|s\rangle$ for which $U\,|0\rangle\,|s\rangle = |0\rangle\,|0\rangle$ and $U\,|1\rangle\,|s\rangle = |1\rangle\,|1\rangle$, and explain why this does *not* let you copy an unknown qubit $\alpha\,|0\rangle + \beta\,|1\rangle$—that is, why it is consistent with the no-cloning theorem.

Problem 2. Quantum teleportation. *(10 pts)* In the teleportation protocol, suppose Penelope's Bell measurement yields $(m_1, m_2) = (1, 0)$. Here the *pre-correction state* means the state of Odysseus's qubit immediately after Penelope's measurement and the transmission of the two classical bits (m_1, m_2), but *before* Odysseus applies any correction gate.

(a) Write down Odysseus's pre-correction state and verify it equals $Z\,|\psi\rangle$.

(b) State the correction Odysseus applies and confirm it restores $|\psi\rangle$.

Problem 3. The reversible two-bit adder. *(10 pts)* Using the half- and full-adder circuits of §1.3.6 of the notes ("Worked example: a reversible binary adder"):

(a) trace the two-bit adder on the input $1 + 2$ (that is, $A_1A_0 = 01$ and $B_1B_0 = 10$) and confirm the measured output is $011_2 = 3$;

(b) count the Toffoli and CNOT gates in the complete two-bit adder.

Part II — Query algorithms and Grover search (Lecture 3)

Problem 4. The four one-bit functions and Deutsch's algorithm. *(10 pts)* List all four functions $f : \{0,1\} \to \{0,1\}$ and classify each as constant or balanced. For each, compute by hand the first-qubit measurement outcome produced by the Deutsch algorithm, and confirm that the outcome distinguishes the constant from the balanced functions in a single query.

Problem 5. Iteration count and over-rotation. *(10 pts)*

(a) For $N = 2^{20}$, approximately how many iterations k^* maximize the success probability? What happens qualitatively to the success probability if one instead runs $2k^*$ iterations? Explain using $p_k = \sin^2((2k+1)\theta/2)$.

(b) For $M = 4$ marked items among $N = 1024$, find $\sin(\theta/2)$, the optimal iteration count k^*, and compare it with the single-solution count for the same N.

Problem 6. Building the phase oracle. *(10 pts)* On three qubits, let the marked state be $x_0 = |101\rangle$. Write the marking reflection $R_{x_0} = \mathbb{I} - 2\,|x_0\rangle\langle x_0|$ as a doubly controlled Z conjugated by X gates, specifying which qubits carry the X gates. Verify by hand that your circuit multiplies $|101\rangle$ by -1 and leaves the other seven basis states fixed (it suffices to check $|101\rangle$, one basis state that becomes $|111\rangle$ under the leading NOTs, and one that does not). Then run *one* Grover iterate on the uniform state $|\psi_0\rangle = H^{\otimes 3}\,|000\rangle$ with this x_0 and report the resulting amplitudes.

Part III — Quantum Fourier transform and phase estimation (Lecture 4)

Problem 7. Exact phase estimation. *(10 pts)* Let $\varphi = 0.011_2 = \frac{3}{8}$ and take $t = 3$ ancilla qubits. Write the ancilla state after the controlled powers in standard QPE, identify it as $\mathrm{QFT}\,|k\rangle$ for the correct k, and conclude the measurement outcome. Why is the readout exact in this case?

Problem 8. How many ancillas? *(10 pts)* Using the precision/qubit-count relation of Lecture 4, determine how many ancilla qubits t are needed to estimate a phase φ to $n = 6$ bits of precision with success probability at least 0.9 (so $\epsilon = 0.1$). State the general formula you use and then evaluate it.

Problems adapted from the exercise sets of Lectures 1, 3, and 4 of the Quantum Scientific Computing course notes.